

IVOMEC® Premix for pigs (Ivermectin)

PRODUCT COMPOSITION: Ivermectin 0.6%w/w

PRODUCT DESCRIPTION: Clean, slightly yellow to brown fine fibrous meal mixture.

PHARMACODYNAMICS: Ivermectin is a member of the macrocyclic lactone class of endectocides which have a unique mode of action. Compounds of the class bind selectively and with high affinity to glutamate-gated chloride ion channels which occur in invertebrate nerve and muscle cells. This leads to an increase in the permeability of the cell membrane to chloride ions with hyperpolarization of the nerve or muscle cell, resulting in paralysis and death of the parasite. Compounds of this class may also interact with other ligand-gated chloride channels, such as those gated by the neurotransmitter gamma-aminobutyric acid (GABA). The margin of safety for compounds of this class is attributable to the fact that mammals do not have glutamate-gated chloride channels, the macrocyclic lactones have a low affinity for other mammalian ligand-gated chloride channels and they do not readily cross the blood-brain barrier.

PHARMACOKINETICS Maximum plasma concentration: After administration of feed containing 2 ppm tritium-labelled ivermectin to swine at the recommended dose rate of 0.1 mg/kg/day, the plasma level of total ivermectin equivalents on-drug was 29.7 ppb. By 21 days off-drug, the mean plasma level was below 0.1 ppb.

Excretion - length of time and route: After administration of feed containing 2 ppm tritium-labelled ivermectin to swine at the recommended dose rate of 0.1 mg/kg/day, liver had the highest on-drug mean total residue level of 237.1 ppb, followed by fat, kidney and muscle at 207.2, 116.8 and 57.5 ppb, respectively. At 3 through 21 days off-drug, fat had the highest mean residue level. By 7 days off-drug mean total residue levels in liver, fat, kidney and muscle were 10.7, 18.0, 3.1 and 2.5 ppb, respectively. The muscle tissue generally contained the least residue. Amountability of dosed radioactivity in excreta collected 7 days on-drug and 21 days off-drug was 95.6 to 105.7%. Only 0.1 to 0.3% of the recovered radioactivity was in urine. The remainder was in the faeces. The majority of radioactivity was excreted by 3 days off-drug.

INDICATIONS: The following parasites can be controlled when IVOMEC Premix for Pigs is incorporated in the finished feed at the recommended rates indicated below.

External parasites - Treatment and control of:
Mange Mites: *Sarcoptes scabiei* var. *suis*
Lice: *Haematopinus suis*

Internal parasites - Treatment and control of:
Gastro-intestinal roundworms:

Ascaris suum - Adults and fourth-stage larvae;
Hyostromylus rubidus - Adults and fourth-stage larvae;
Oesophagostomum spp. - Adults and fourth-stage larvae;
Strongylus edentatus - Adults
Lungworms: *Metastrongylus* spp. - Adults

* Given to pregnant sows before farrowing, it effectively controls transmission via milk of *S. ransomi* to piglets

DOSEAGE AND ADMINISTRATION: The recommended dose level is 100 mcg ivermectin/kg bodyweight fed daily for seven consecutive days. The appropriate inclusion rate of the premix, in kilograms per tonne of finished feed, can be calculated as follows:

$$\text{Premix inclusion rate (kg/tonne feed)} = \frac{0.1 \times \text{average bodyweight (kg)}}{6 \times \text{average daily feed intake (kg)}}$$

Growing Pigs: The recommended dose level of 100 mcg/kg bodyweight daily for seven days is obtained under most circumstances, for pigs up to 40 kg bodyweight, by including 333g of IVOMEC Premix in each metric tonne of final feed. The IVOMEC Premix should be thoroughly mixed in the finished feed and fed continuously as the only ration for seven consecutive days. In pigs weighing 40 kg liveweight and over, average daily feed consumption may fall below a feed intake of 5% where restricted feeding programmes are in use or where pigs are fed a ration high in protein. For pigs weighing 40 kg and over include 400 g of IVOMEC Premix in each metric tonne of final feed.

Adult Pigs: The recommended dose level for adult pigs weighing over 100 kg liveweight is achieved under most circumstances by thoroughly mixing 1.67 kg of IVOMEC Premix with 1 tonne of pig feed. The resultant medicated feed is to be fed at the rate of 1 kg per 100 kg of bodyweight each day for seven consecutive days, as part of the individual ration. Where medicated feed is to be fed as part of the ration, it is recommended that the feed medicated with IVOMEC Premix is fed first. After this is consumed, any remainder of the daily feed allocation should be provided. This should be repeated for seven consecutive days.

Alternatively, where dry feed intake can be accurately determined and

all animals to be treated have similar bodyweight, the inclusion rate can be calculated using the previous formula to allow sole feeding of medicated feed.

RECOMMENDED TREATMENT PROGRAMME

Growing Pigs: Groups of growing pigs should be treated by feeding medicated feed for seven consecutive days on transfer to clean quarters. Where an all-in-all-out system is not possible, it is recommended that IVOMEC in-feed parasite control programme should begin with treatment of all growing pigs already in the house.

Breeding Animals: Breeding animals are treated by feeding medicated feed for seven consecutive days. At the time of initiating any parasite control programme, it is important to treat all animals in the herd. After the initial treatment, use IVOMEC Premix regularly as follows:

Sows: Treat, preferably 14-21 days prior to farrowing, to minimize infection of piglets.

Glts: Treat 14-21 days prior to service. Treat 14-21 days prior to farrowing.

Boars: Treat at least 2 times per year. Frequency of and need for treatments are dependent upon parasite exposure.

Note (1): Exposure of treated pigs to infected animals, contaminated premises, soil or pasture may result in reinfestation and retreatment may be necessary.

Note (2): Since the effect of ivermectin on mange mites is not immediate, avoid direct contact between treated and untreated pigs for at least one week after completion of treatment.

Note (3): Since louse eggs are unaffected by ivermectin and may take up to three weeks to hatch, retreatment may be necessary.

CONTRA-INDICATIONS, WARNINGS, ETC.: Animals must not be slaughtered for human consumption during treatment. Pigs weighing less than 100kg may be slaughtered for human consumption only after 3 days from the last treatment. Pigs weighing more than 100kg may be slaughtered for human consumption only after 12 days from the last treatment. Do not smoke or eat while handling the product. Wash hands after use. Mixing of the product with feed must take place in a well ventilated area. Avoid contact with skin and eyes. In case of accidental contact, wash the affected area thoroughly with clean running water. If eye irritation persists, seek medical advice. For animal treatment only. Keep out of the reach of children. Do not use for any other animal species as severe adverse reactions, including fatalities in dogs, may occur.

USE DURING PREGNANCY AND LACTATION: At the recommended use level, no adverse effects on fertility or gestation in breeding animals were observed.

INTERACTION WITH OTHER MEDICAMENTS: No incompatibilities with other commonly used products were observed.

SIDE EFFECTS: No undesirable effects have been observed when the product is administered to swine at the recommended dose rate.

OVERDOSE: When included in the ration of pigs at levels up to 5 times the recommended dose of 0.1 mg ivermectin per kg bodyweight for 21 consecutive days (3 times the recommended treatment period), the product did not produce treatment related adverse reactions. No antidote has been identified.

ENVIRONMENTAL SAFETY

EXTREMELY DANGEROUS TO FISH AND AQUATIC LIFE. Do not contaminate surface water or ditches with the product or used containers. Drug containers and any residual contents should be disposed of safely, e.g., by burying in waste ground away from water courses or incinerated.

PHARMACEUTICAL WARNINGS: To ensure thorough dispersion of the product, it should first be mixed with a suitable quantity of feed ingredients before incorporation in the final mix. The product will remain stable in the finished feed for three months when stored below 25° C. The product can be incorporated in pelleted feed preconditioned with steam for up to 10 seconds at a temperature not exceeding 85° C.

Do not store above 25°C. Store in dry place.

Shelf life in finished feed: 3 months

WITHDRAWAL PERIODS: Pigs (over 100kg): 12 days; Pigs (under 100kg): 3 days

Packaging: 5kg multiwall paper bags with polyethylene inner liner
Manufactured by:

DOPHARMA

Zalmweg 24, 4941 VX Raamsdonksveer, The Netherlands
For: Boehringer Ingelheim

Product Registration Holder

AgriTech Enterprise Sdn Bhd

No 15&17, Jalan Teknologi 3/5A, Pusat Teknologi Sinar Damansara, PJUS, Kota Damansara, 47810 Petaling Jaya, Selangor Darul Ehsan
Reg No: MAL12020007HAR

• IVOMEC is registered trademark of Boehringer Ingelheim

Expiry Date:

Manufacturing date:

Batch No.:

Code DOPHARMA

LABEL CODE:	VERSION	COLORS
5LIVOME3-05K PRODUCT: IVOMEC PREMIX 0.6%W/W PACK SIZE: 5 KG COUNTRY: MY DIMENSIONS: cm = 14.7 x 20	Version: 1 MMM/YY: 27/5/2020	BLACK ! Please use the official PANTONE® (P) matching system for an accurate color representation.

Marketing approval:	Regulatory approval:
COMPLETE NAME: DATE + SIGNATURE:	COMPLETE NAME: DATE + SIGNATURE: